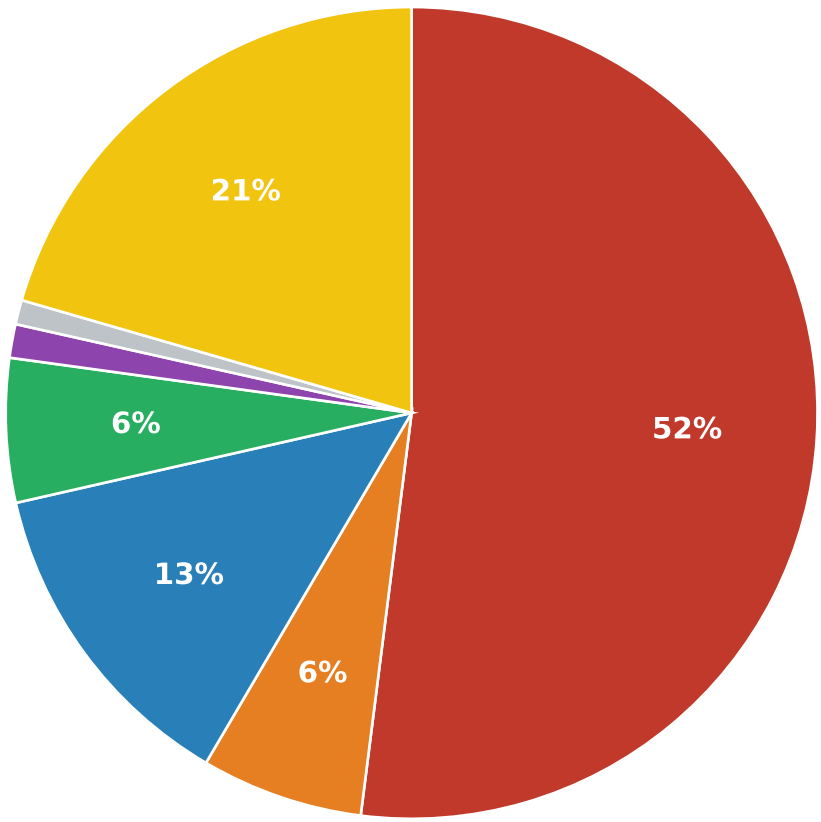
SpeechNet, one update step on GAP9 (GVSoC), board L1/L2 settings: what occupies L2 at the peak

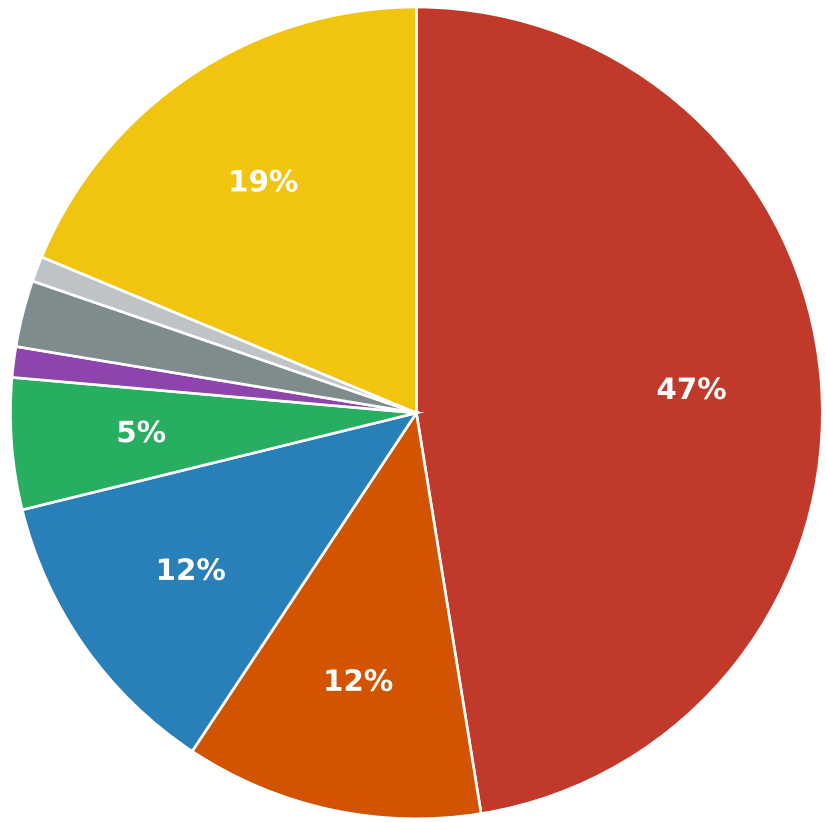
QZO+ECO (fused, cluster)
L2 peak 301,960 B (schedule step 18)



Block-0 activations: 157,024 B (52.0%)
Conv input (HWC copy): 19,488 B (6.5%)
Data window + label: 39,208 B (13.0%)
Trainable parameters (master): 17,316 B (5.7%)
Perturbed parameter copies: 4,040 B (1.3%)
Static constants: 2,928 B (1.0%)
ECO momentum (harness): 61,956 B (20.5%)

- Block-0 activations: Conv0 output [A+ | A-], int8 14x701x16, HWC (157,024 B)
- Conv input (HWC copy): block 1's conv input [x+ | x-], int8 14x87x16, HWC (19,488 B)
- Data window + label: data window, fp32 14x700 (39,200 B); label (8 B)
- Trainable parameters (master): 22 tensors = the graph's trainable inputs (input_2..23): conv W and b, BN gamma and beta of blocks 0-4, FC W and b; live for the whole step (the update graph changes them in place)
- Perturbed parameter copies: 14 tensors: BN gamma/beta of blocks 0-4 as stacked [+; -] copies and the FC W/b as + and - copies; the conv weights are perturbed inside the fused conv kernels (L1 scratch), so they have no L2 copy
- Static constants: 26 compile-time constant tensors (multipliers, shifts)
- ECO momentum (harness): 22 fp32 buffers, one value per trainable parameter (15,489), allocated by the C harness, 61,956 B

QZO+ECO (fused, NE16 blocks 0-2)
L2 peak 330,912 B (schedule step 23)



Block-0 activations: 157,024 B (47.5%)
im2col column (NE16): 39,256 B (11.9%)
Data window + label: 39,208 B (11.8%)
Trainable parameters (master): 17,316 B (5.2%)
Perturbed parameter copies: 4,040 B (1.2%)
NE16 encoded weights + bias: 8,768 B (2.6%)
Static constants: 3,344 B (1.0%)
ECO momentum (harness): 61,956 B (18.7%)

- Block-0 activations: Conv0 output [A+ | A-], int8 14x701x16, HWC (157,024 B)
- im2col column (NE16): block 0's im2col column: 9,814 output pixels x K = 4 bytes, built by the cluster for the NE16 1x1 conv (39,256 B; its Reshape view is the same buffer)
- Data window + label: data window, fp32 14x700 (39,200 B); label (8 B)
- Trainable parameters (master): 22 tensors = the graph's trainable inputs (input_2..23): conv W and b, BN gamma and beta of blocks 0-4, FC W and b; live for the whole step (the update graph changes them in place)
- Perturbed parameter copies: 14 tensors: BN gamma/beta of blocks 0-4 as stacked [+; -] copies and the FC W/b as + and - copies; the conv weights are perturbed inside the fused conv kernels (L1 scratch), so they have no L2 copy
- NE16 encoded weights + bias: 6 tensors: [W+; W-] of blocks 0-2 in NE16 bit-plane format (block 1, 2: 4,096 B each; block 0: 256 B) + their corrected requant bias; prepared at the start of the schedule, 8,768 B
- Static constants: 34 compile-time constant tensors (multipliers, shifts)
- ECO momentum (harness): 22 fp32 buffers, one value per trainable parameter (15,489), allocated by the C harness, 61,956 B